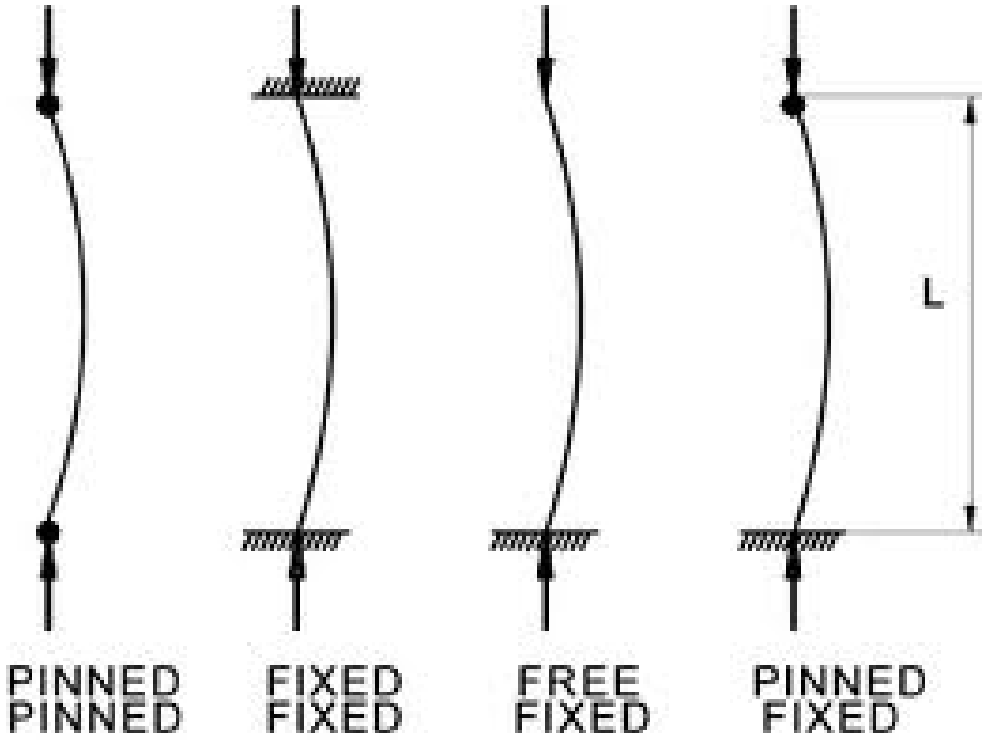
STRUT

A strut is a long thin compression member. It may collapse under a compressive load by buckling and bowing out.



EULER CRIPPLING LOAD

Definition of Strut

- A long structural member, subjected to an axial compressive force, is called a strut. It may be horizontal, inclined or even vertical.

- But a short vertical member subjected to an axial compressive force, used in buildings or frames is called a *column*.

- When a column or a strut is subjected to some compressive force, then a compressive stress is induced,

$$\sigma = \frac{p}{A}$$

where P Compressive force and

A Cross-sectional area of the column

EULER CRIPPLING LOAD

- It was observed that if the force or load is gradually increased the column will reach a stage, when it will be subjected to the ultimate crushing stress.

- Beyond this stage, the column will fail by crushing. The load corresponding to the crushing stress, is called *crushing load.*

- It has also been observed that in the case of long columns, a compression member does not fail entirely by crushing, but by bending *i.e.*, ***buckling.***

Euler's Column Theory

- The first rational attempt, to study the stability of long columns, was made by Mr. Euler.
- He derived an equation, for the buckling load of long columns based on the bending stress.
- While deriving this equation, the effect of direct stress is neglected.
- This may be justified with the statement that the direct stress induced in a long column is negligible as compared to the bending stress.
- It may be noted that the Euler's formula cannot be used in the case of short columns, because the direct stress is considerable and hence cannot be neglected.

Assumptions in the Euler's Column Theory

The following simplifying assumptions are made in the Euler's column theory:

- Initially the column is perfectly straight and the load applied is truly axial.
- The cross-section of the column is uniform throughout its length.
- The column material is perfectly elastic, homogeneous and isotropic and thus obeys Hooke’s law
- The length of column is very large as compared to its cross-sectional dimensions.
- The shortening of column, due to direct compression (being very small) is neglected.
- The failure of column occurs due to buckling alone

SLENDERNESS RATIO

A strut is usually defined by its slenderness ratio. Which is defined as,

S.R. = $\frac{L}{k}$

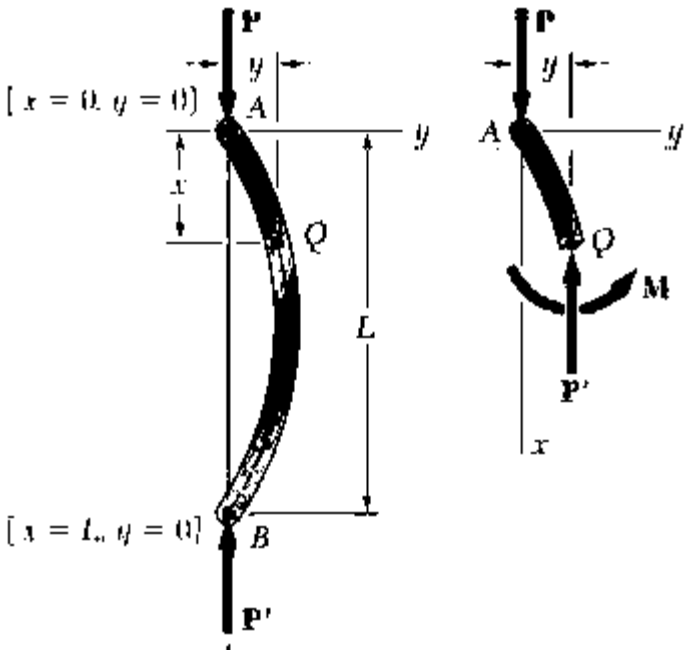
- L is the effective length and k is the **least** radius of gyration for the cross sectional area.
- For steel materials, struts are members with slenderness ratio exceeding 100.

Example; Derive formulae for the radius of gyration of a shaft with a circular cross-section of diameter D and another with a rectangular cross-section of width B and depth D.

Circle; $I = \frac{\pi D^4}{64}$; $I = Ak^2 \implies k = \sqrt{\frac{D^2}{16}} = \frac{D}{4}$

Rectangle; $I = \frac{BD^3}{12}$; $I = Ak^2 \implies k = \sqrt{\frac{D^2}{12}} = \frac{D}{2\sqrt{3}}$

Case 1: Critical Load
Pin-ended (hinged) strut



$$\therefore EI\frac{d^2y}{dx^2} = M = -Py$$

$$EI\frac{d^2y}{dx^2} + Py = 0 \text{ or } \frac{d^2y}{dx^2} + \frac{P}{EI}y = 0$$

$$\frac{d^2y}{dx^2} + \alpha^2 y = 0; \text{ where } \alpha^2 = \frac{P}{EI}$$

$$y = A\sin\alpha x + B\cos\alpha x$$

A and B can be obtained from the boundary conditions

Case 1: Critical Load Pin-ended (hinged) strut cont'd…

$@\ x = 0, y = 0 \Rightarrow B = 0;$

$@\ x = l, y = 0 \Rightarrow A\sin\alpha l = 0$

The least value of αl to satisfy this equation, is $\alpha l = n\pi, n = 0, 1, 2, 3...$

$\therefore$ the least value of P to cause the strut to buckle (Euler crippling load) is

$$\mathrm{P_e} = EI\left(\frac{\pi}{l}\right)^2$$

Example 1

A straight bar of alloy, 1 m long and 12·5 mm by 4·8 mm in section, is mounted in a strut-testing machine and loaded axially until it buckles. Assuming the Euler formula to apply, estimate the maximum central deflection before the material attains its yield point of 280 N/mm^2.

E = 72,000 N/mm^2.

Example 1 cont'd…

There will be no deflection at all until the Euler load is reached, i.e,

$$P_e = EI\left(\frac{\pi}{l}\right)^2 ; I = \frac{12.5x4.8^3}{12} = 115.2\ mm^4$$

$$P_e = 72\,000\ x\ 115.2\left(\frac{\pi}{1000}\right)^2 = 82\ N$$

$$280 = \frac{82}{12.5x4.8} + \frac{82\delta\ x2.4}{115.2}$$

$$\delta = 163\ mm$$

Assuming the deflection at the mid - point is δ.

Then Maximum bending moment is $P_e\delta$,

maximum stress is the sum of direct and bending

stresses at the centre. That is $\sigma = \frac{F}{A} + \frac{Mc}{I}$

Example 2

A uniform bar of cross-sectional area A and flexural stiffness EI is heated so that its temperature varies linearly from ½t at one end to t at the other end. One end is pin-jointed to a rigid foundation; the other end is pin-jointed so that it can slide in the direction of the length of the bar, the thermal expansion of which is resisted by a compression spring of stiffness k.

If there is no load in the spring when t =0, obtain an expression for the stress in the bar when it is heated and show that it buckles in flexure when

$$t = \frac{4\pi^2 I}{3\alpha l^2 A}\left(1 + \frac{EA}{kl}\right)$$

where α=coefficient of linear thermal expansion.

Example 2 cont'd…

- The average temperature along the bar is *¾t,* and hence the thermal expansion of the bar is *¾ αlt.*
- If *P* is the force exerted by the spring on the bar, the compression produced is, *Pl/AE* and the compression of the spring is *P/k.*
- Net expansion of bar =compression of spring, that is

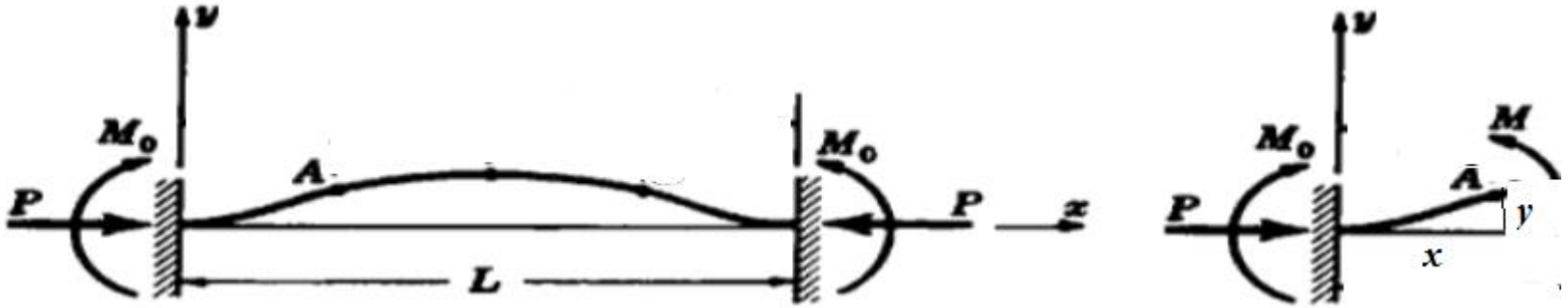
$$¾\,\alpha lt - Pl/AE = P/k$$

$$P = \frac{\frac{3}{4}\alpha lt}{k + l/AE} = EI\left(\frac{\pi}{l}\right)^2$$

From which

$$t = \frac{4\pi^2 I}{3\alpha l^2 A}\left(1 + \frac{EA}{kl}\right)$$

Case 2: Strut Fixed at Both Ends



Suppose the strut to have deflected, and let M_o be the end fixing moment

$$\Rightarrow y = \left(\frac{M_o}{P}\right)(1-\cos\alpha x)$$

$$EI\, d^2y/dx^2 = -Py + M_o$$

$$\frac{d^2y}{dx^2} + \alpha^2 y = 0$$

$$y = A\sin\alpha x + B\cos\alpha x + M_o/EI\alpha^2$$

$$x = 0; \quad y = 0; \quad \therefore B = -M_o/EI\alpha^2 = -M_o/P$$

$$x = 0; \quad dy/dx = 0; \quad \therefore A = 0$$

$$x = l; \quad y = 0; \quad \therefore \cos\alpha l = 1$$

$$\alpha l = 2\pi$$

The least solution is

The buckling load, $P_e = 4\pi^2 EI/l^2$

Example 3

A strut of length 2a has each end fixed in an elastic material which exerts a restraining moment per unit angular displacement(μ). Prove that the critical load P is given by the equation , μn tan (na)+P=0;

Where n^2=P/EI . Such a strut, 2·5 m in length, has a theoretical critical load of 15 kN on the assumption of pinned ends. Determine the percentage increase in the critical load if the constraint offered at the ends is 170 Nm per degree of rotation.

Example 3 cont'd…

- *Let M be the restraining moment at each end, then the general solution is obtained as,* $y=A\sin nx+B\cos nx+M/EIn^2$ *using the notation given in the question.*
- *@* $x=0$, $y=0$ *and* $B=-M/P$
- *Also, @*$x=0$, $(dy/dx)= M/\mu$
- $dy/dx=nA\cos nx-Bn\sin nx$
- $\Longrightarrow A=M/\mu n$
- *At the center,* $x=a$, $dy/dx=0$
- *That is* $\tan na=A/B= (M/\mu n)(-P/M)=-P/ \mu n$
- $\mu n \tan (na)+P=0$;

Example 3 cont'd…

- *For pinned ends,*
- $P_e = \pi^2 EI/(2a)^2 = 15\,000 \Longrightarrow EI = 9\,500\ Nm^2$

From $n^2 = P/EI$ *or* $n = \sqrt{P}/97.5$

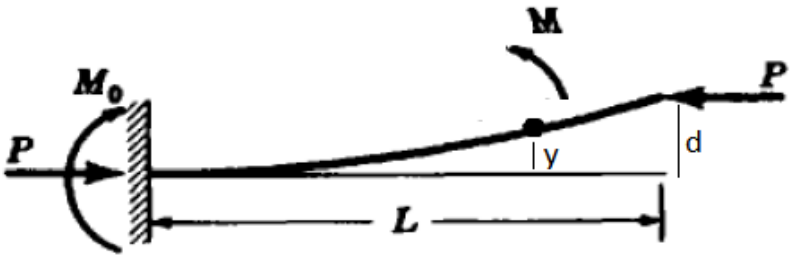
and $\mu = 170/1^o = 170/(\pi/180) = 9740.28\ Nm/rad$

- *Therefore* $\tan(0.0128\sqrt{P}) = -0.01P$ *, which gives*
- $\sqrt{P} = 166$ *or* $P = 27.5\ kN$

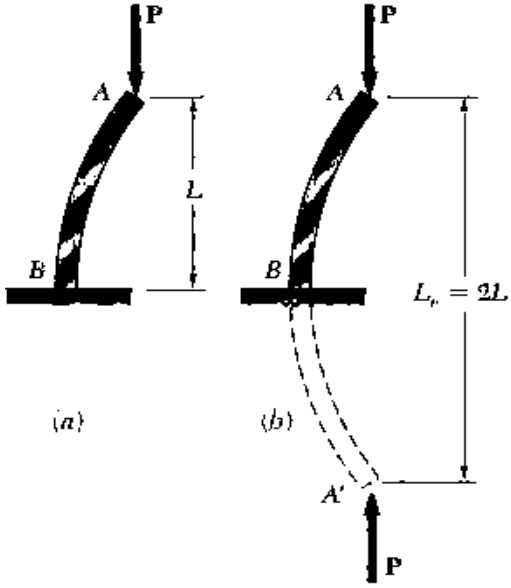
Percentage increase in the critical load,e

e=(27.5-15)/15=83.33%

Case 3: Strut with Pin-joint at one end and free at the other end







$$M + P(d - y) = 0$$

$$EI \, d^2y / dx^2 = -P(y - d)$$

$$y = A\sin \alpha x + B\cos \alpha x + d$$

$$@\ x = 0, y = 0 \Rightarrow B = -d$$

$$@\ x = 0, \frac{dy}{dx} = 0 \therefore A = 0$$

$$@\ x = l, y = d$$

$$d = d(1 - \cos \alpha l) \Rightarrow$$

$$\alpha l = n\pi / 2, n = 1,3,5...$$

the minimum force is

$$P = EI\left(\frac{\pi}{2l}\right)^2$$

Case 4: Strut with Pin-joint at one end(A) and Fixed at the other end(B)



$$M + Py - Vx = 0$$

$$EI \, d^2y/dx^2 = -Py + Vx$$

$$\frac{d^2y}{dx^2} + \frac{P}{EI} y = \frac{V}{EI} x$$

$$y = A\sin \alpha x + B\cos \alpha x + (V/P)x$$

$$@\ x = 0, y = 0; B = 0$$

$$@\ x = l, y = 0, A = -\frac{(V/P)l}{\sin \alpha l}$$

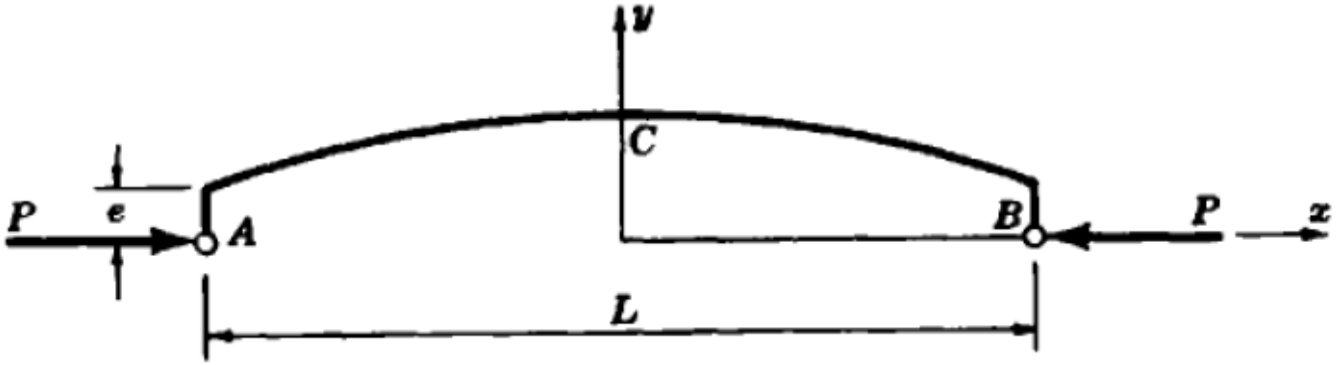
$$@\ x = l, \frac{dy}{dx} = 0; \frac{dy}{dx} = A\alpha \cos \alpha x + V/P$$

$$\tan \alpha l = \alpha l = 4.493$$

$$\Rightarrow \alpha = 4.493/l$$

$$Since\ \alpha^2 = \frac{P}{EI} \Rightarrow P = \alpha^2 EI = \frac{2.047\pi^2 EI}{L^2}$$

Case 5: Strut with Eccentric Load



$$\therefore EI\frac{dy^2}{dx^2} = M = -Py$$

$$y = A\sin\alpha x + B\cos\alpha x$$

$$@\ x = 0, y = e; B = e$$

$$@\ x = L/2, dy/dx = 0$$

$$A = e\tan\frac{\alpha L}{2}$$

$$y = e(\tan\frac{\alpha L}{2}\sin\alpha x + \cos\alpha x)$$

The deflection becomes infinite for

$$\tan\alpha l/2 = \infty; \alpha l = \pi$$

$$P_e = \pi^2 EI/l^2$$

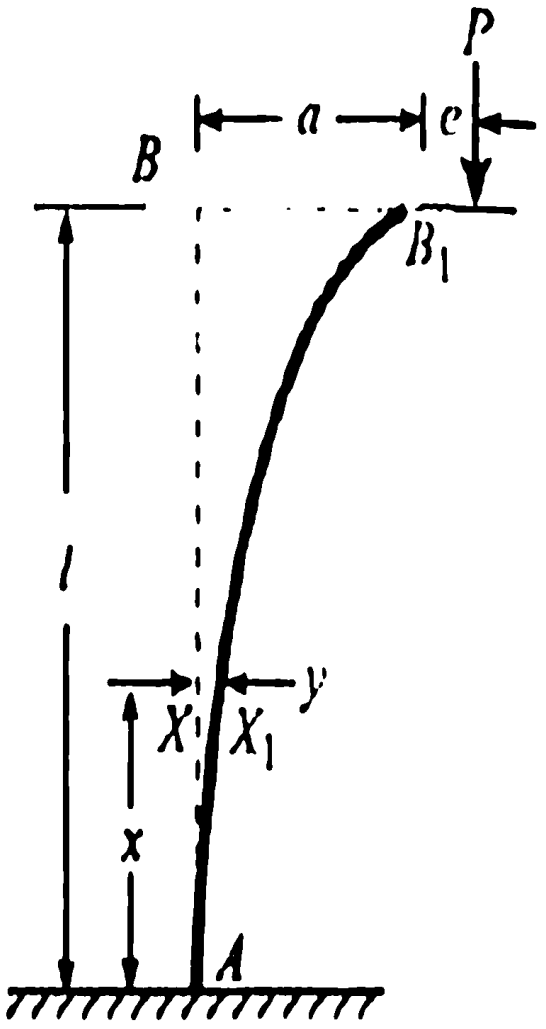
$$@\ centre$$

$$\hat{y} = e\left[\left(\tan\alpha l/2\right)\sin\alpha l/2 + \cos\alpha l/2\right]$$

$$= e\left(\frac{\sin^2\alpha l/2 + \cos^2\alpha l/2}{\cos\alpha l/2}\right) = e\sec\alpha l/2$$

Case 5_2: Euler's formula

This case could be looked at as case 5 with equivalent length of $2l$, since $a+e$ gives the maximum deflection in this member.



$$P_e = \pi^2 EI \Big/ (2l)^2$$

@ point A

$$\hat{y} = a + e = e\sec\alpha(2 \times l/2)$$

$$\hat{y} = e\sec\alpha l$$

Example

An alloy hollow circular column of 200 mm external and 160 mm internal diameter is 5 m long and fixed at both of its ends. It is subjected to a load of 120 kN at an eccentricity of 20 mm from the geometrical axis. Determine the maximum stress induced in the column section. Take E as 120 GPa.

- **Solution**
- Given: External diameter *(D)* = 200 mm; Internal diameter (*d*) = 160 mm; Length *(l)* = 5 m = 5 x 10^3 mm; Load *(P)* = 120 kN = 120 x 10^3 N; Eccentricity *(e)* = 20 mm and modulus of elasticity (*E*) = 120 GPa = 120 x 10^3 N/mm^2

- Area of the column section

$$A = \frac{\pi}{4}\left[D^2 - d^2\right] = \frac{\pi}{4}\left[(200)^2 - (160)^2\right] = 11.31\times10^3\ mm^2$$

- Moment of inertia of column section

$$I = \frac{\pi}{64}\left[D^4 - d^4\right] = \frac{\pi}{64}\left[(200)^4 - (160)^4\right] = 46.37\times10^6\ mm^4$$

- Modulus of section

$$Z = \frac{I}{D/2} = \frac{46.37x10^6}{200/2} = 463.7\times10^3\ mm^3$$

- Since the column is fixed at its both ends, the maximum deflection is

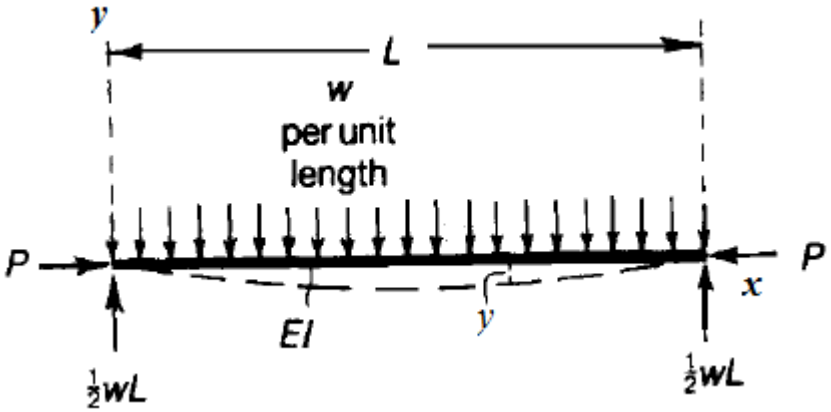
$$\hat{y} = e\sec\alpha(l/2)$$

$$\alpha l/2 = \left(\frac{l}{2}\sqrt{\frac{P}{EI}}\right) = \frac{5000}{2}\sqrt{\frac{120\times10^3}{(120\times10^3)(46.37\times10^6)}} = 0.367\ rad = 21.04^o$$

- The maximum compressive stress induced in the column

$$\sigma = \frac{P}{A} + \frac{1}{Z}.P.e.\sec\left(\frac{l}{2}\sqrt{\frac{P}{EI}}\right) = \frac{120x10^3}{11.31x10^3} + \frac{(120x10^3)(20\sec 21.04^0)}{463.7x10^3} = 15.12\ N/mm^2$$

Case 6: Pin-jointed Strut with Distributed Load



The bending moment is given by

$$M = \frac{wl}{2}x - \frac{w}{2}x^2 + Py$$

$$\Rightarrow EI\frac{d^2y}{dx^2} = -\left[\frac{wl}{2}x - \frac{w}{2}x^2 + Py\right]$$

$$y = A\sin\alpha x + B\cos\alpha x + \frac{wx^2}{2P} - \frac{wl}{2P}x - \frac{w}{\alpha^2 P}$$

$$@\ x = 0, y = 0$$

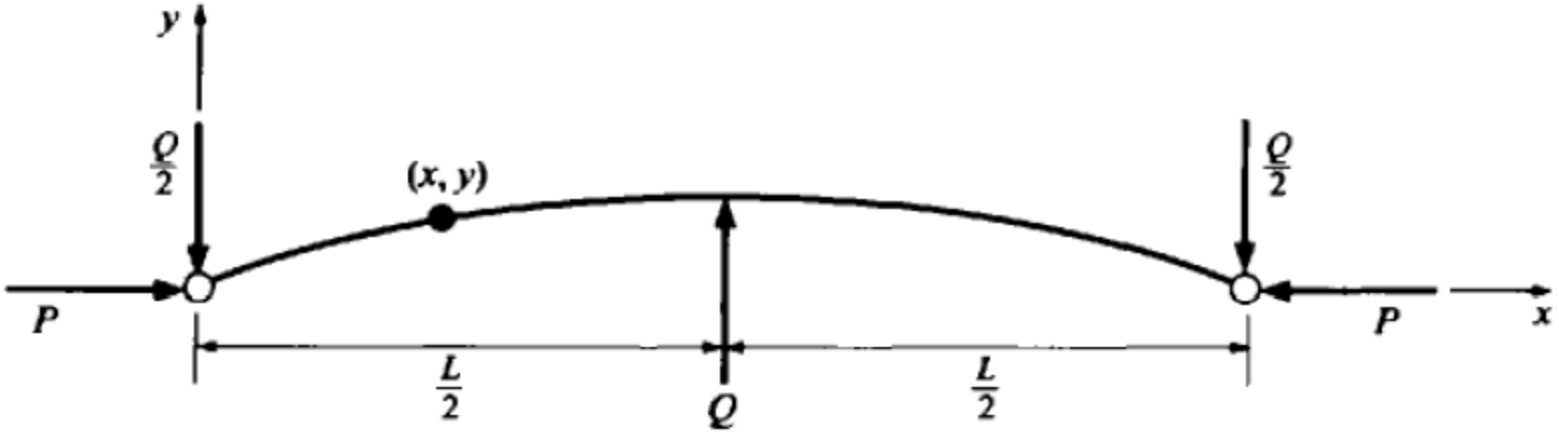
$$B = \frac{w}{\alpha^2 P}$$

$$@\ x = l, y = 0; \therefore A = \frac{w}{\alpha^2 P}\left(\frac{1-\cos\alpha l}{\sin\alpha l}\right)$$

$$@\ y_{\max} = \delta, x = l/2$$

$$\delta = \frac{w}{\alpha^2 P}\left[\sec\frac{\alpha l}{2} - 1\right] - \frac{wl^2}{8P}$$

Case 7: Pin –jointed Strut subjected to a Concentrated Load W at the middle



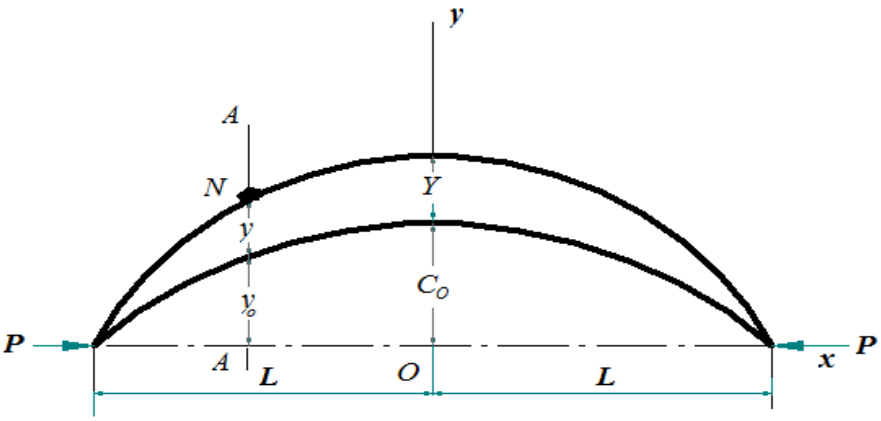
Assignment(*Discuss*)

Show that maximum deflection is given by

$$\delta = \frac{Q}{2P\alpha}\tan\left(\frac{\alpha l}{2}\right) - \frac{Ql}{4P}$$

Perry- Robertson Formula

This is based on the assured initial imperfection in a strut which can be represented as showing a slight initial curvature. The bent shape is assumed to be sinusoidal.



The shape of the initial strut is

$$y_0 = C_0.\cos(\pi x/L)$$

Note when $x = 0$ (at the middle); $y_O = C_0$;

$x = \pm L$, (at the ends); $y_0 = 0$.

Let y be an additional defection due to load P which attains a maximum value Y at $x = 0$.

Bending Moment at point N on AA :

$$M_{AA} = -P(y_0 + y) = -P(y + C_0.\cos \pi x/L)$$

Perry- Robertson Formula cont'd…

$$\therefore EI\frac{d^2y}{dx^2} + Py = -PC_0.\cos \pi x/L$$

$$y = A\sin \alpha x + B\cos \alpha x - \left[\frac{\alpha^2 C_0}{\left(-\pi^2/L^2\right)+\alpha^2}\right]\cos\frac{\pi x}{L}$$

$$x = \pm L;\ y = 0; \quad \therefore A = B = 0$$

$$y = -\left[\frac{\alpha^2 C_0}{\left(-\pi^2/L^2\right)+\alpha^2}\right]\cos\frac{\pi x}{L}$$

$$\text{At } x = 0,\ y = Y$$

$$\text{Thus} \quad Y = -\left[\frac{\alpha^2 C_0}{\left(-\pi^2/L^2\right)+\alpha^2}\right] = -\frac{PC_0}{P - EI\left(\pi/L\right)^2}$$

$$we \text{ know; } P_e = \frac{\pi^2}{L^2}EI \Rightarrow Y = \frac{PC_0}{P_e - P}$$

Perry- Robertson Formula cont'd…

Total maximum deflection is

$$Y_{\max} = C_0 + Y = C_0\left(1 + \frac{P}{P_e - P}\right) = \frac{C_0 P_e}{P_e - P}$$

The maximum compression stress

$$\sigma_{\max} = \frac{P}{A} + \frac{M_{\max}}{I}\upsilon = \frac{P}{A} + \frac{PY_{\max}}{I}\nu$$

$$I = Ak^2$$

$$\sigma_{\max} = \frac{P}{A} + \frac{C_0 P P_e}{\left(P_e - P\right)} \cdot \frac{\nu}{Ak^2}$$

$$\sigma_{\max} = \frac{P}{A} + \frac{C_0\left(P/A\right)\left(P_e/A\right)}{\left(\left(P_e/A\right) - \left(P/A\right)\right)} \cdot \frac{\nu}{\left(A/A\right)k^2}$$

$$\sigma_{\max} = \sigma + \frac{C_0 \sigma \sigma_e}{\left(\sigma_e - \sigma\right)} \cdot \frac{\nu}{k^2}$$

Example

A steel strut has an outside diameter of 180mm and inside diameter of 120mm and is 6m long. It is hinged at both ends and is initially bent. Assuming the centre line of the strut as sinusoidal with maximum deviation of 9mm, determine the maximum stress developed due to an axial load of 150kN, take E=208 GPa

Solution

- Given: Outside diameter of the strut, (D) = 180 mm; Inside diameter of the strut, (d) = 120 mm; Length of the strut, (l) = 6 m =6 x 10^3 mm; Maximum deviation at the centre, (δ') = 9 mm; Young's modulus, (E) = 208 GPa = 208 x 10^3 N/mm^2; Axial load, (P) = 150 kN = 150 x 10^3 N

- *Maximum stress developed*

- Area of cross-section $A = \frac{\pi}{4}\left(D^2 - d^2\right) = \frac{\pi}{4}\left(180^2 - 120^2\right) = 14.14x10^3\ mm^2$

- Moment of inertia $I = \frac{\pi}{64}\left(D^4 - d^4\right) = \frac{\pi}{64}\left(180^4 - 120^4\right) = 41.35x10^6\ mm^4$

- Radius of gyration $k^2 = \frac{I}{A} = \frac{41.35x10^6\ mm^4}{14.14x10^3\ mm^2} = 2.924x10^3\ mm^2$

- Euler load, for pinned at both ends, $L_e = l = 6x10^3$ mm
- Therefore $$P_E = \left(\frac{\pi}{L_e}\right)^2 EI = \left(\frac{\pi}{6x10^3}\right)^2 \left(208x10^3\right)\left(41.35x10^6\right) = 2.36x10^6 \ N$$
- Euler Stress $$\sigma_e = \frac{P_E}{A} = \frac{2.36x10^6}{14.14x10^3} = 166.75 \ N/mm^2$$
- Direct stress $$\sigma = \frac{P}{A} = \frac{150x10^3}{14.14x10^3} = 10.6 \ N/mm^2$$
- Distance of the extreme layer in compression from the neutral axis $$y = \frac{D}{2} = \frac{180}{2} = 90 \ mm$$
- We know that $$\sigma_{\max} = \sigma + \frac{C_0 \sigma \sigma_e}{\left(\sigma_e - \sigma\right)} . \frac{y}{k^2} = 10.6 + \frac{9 \times 10.6 \times 166.75}{166.75 - 10.6} . \frac{90}{2924}$$
- Therefore $$\sigma_{\max} = 3.14 \ N/mm^2$$

Rankine Formula

For a very short columns failure is by crushing (or yielding)

$P_c = A.\sigma_c$; σ_c is the *crushing stress*

On the other hand for very long struts the

failure will occur through buckling as in Euler load

$$P_e = \frac{c\pi^2 EI}{l^2}$$

[where c = 1,2,4,1/4 depending on end conditions]

If P_R is the Rankine load for the failure of any

length of strut then $\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_e}$.

Let us examine this equation for its validity.

For a very short column, P_e is large, thus

$$\frac{1}{P_e} \approx small$$

Rankine Formula cont'd…

This implies

$$\frac{1}{P_R} \cong \frac{1}{P_c} \quad \therefore P_R \cong P_c$$

For a very long column P_e is small, thus

$$\frac{1}{P_e} \approx \text{large}$$

This implies; $\frac{1}{P_R} \cong \frac{1}{P_e} \quad \therefore P_R \cong P_e$

Rewriting ; $\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_e} = \frac{P_e + P_c}{P_c P_e}$

Thus $P_R = \frac{P_c P_e}{P_e + P_c} = \frac{P_c}{1 + P_c / P_e} = \frac{A\sigma_c}{1 + a(l/k)^2}$

Where $a = \frac{\sigma_c}{c\pi^2 E}$

- The following table gives the values of crushing stress (σ_c) and Rankine's constant (a) for various materials:

S.No.	Material	σ_C in MPa	$a = \frac{\sigma_C}{\pi^2 E}$
1.	Mild Steel	320	$\frac{1}{7500}$
2.	Cast Iron	550	$\frac{1}{1600}$
3.	Wrought Iron	250	$\frac{1}{9000}$
4.	Timber	40	$\frac{1}{750}$

- **Note : The above values arc only for a column with both ends hinged. For other end conditions, the equivalent length should be used.**

- ***Example 2-8****. Find the Euler's crippling load for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thick. Take length of the column as 2.3 m and hinged at its both ends. Take E = 205 GPa. Also determine crippling load by Rankine's formula using constants* σ_c *and* a *as 335 MPa and 1/7500 respectively.*

- **Solution**
- Give: External diameter (*D*) = 38 mm; Thickness = 2.5 mm or inner diameter (*d*) = 38- (2 x 2.5) = 33 mm ; Length of the column (*1*) = 2.3 m = 2.3 x 10^3 mm; Yield stress (σ_c) = 335 MPa = 335 N/mm^2 and Rankine's constant (a) =1/7500
- *Euler's crippling load*
- We know that moment of inertia of the column section

$$I_{XX} = \frac{\pi}{64}\left(D^4 - d^4\right) = \frac{\pi}{64}\left[(38)^4 - (33)^4\right] = 14.05x10^3\,\pi\ mm^4$$

- Since the column is hinged at its both ends, therefore effective length of the column, $L_e = l = 2.3 \times 10^3$ mm

- Therefore Euler's crippling load

$$P_E = \pi^2 EI \Big/ L_e^2 = \frac{\pi^2 \left(205x10^3\right)\left(14.05x10^3\pi\right)}{\left(2300\right)^2} = 16{,}880\ N$$

- *Rankine's crippling load*
- We know that area of the column section

$$A = \frac{\pi}{4}\left(D^2 - d^2\right) = \frac{\pi}{4}\left[\left(38\right)^2 - \left(33\right)^2\right] = 88.75\pi\ mm^2$$

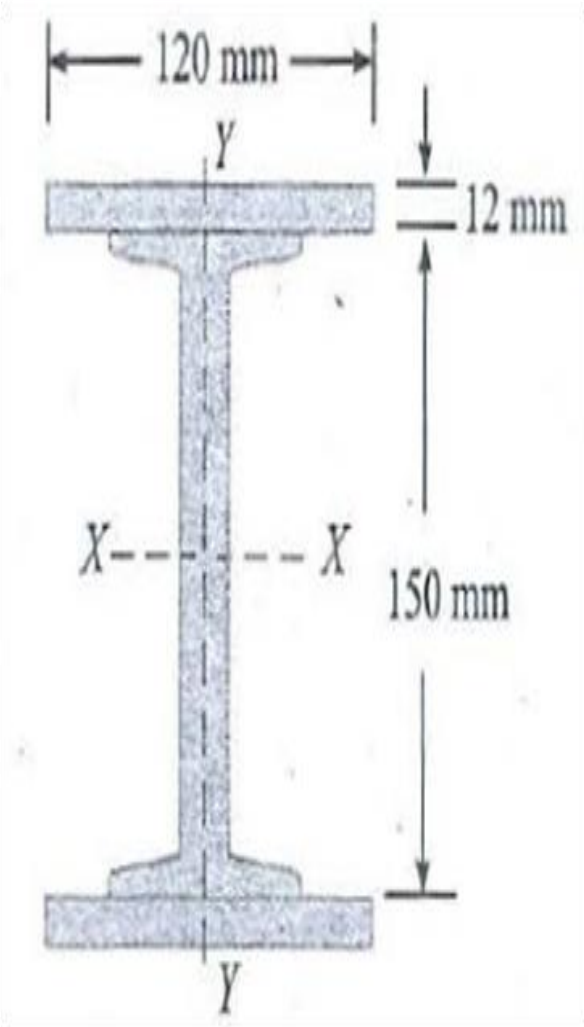
- The least radius of gyration

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{14.05x10^3\pi}{88.75\pi}} = 12.6\ mm$$

- Therefore, Rankine's crippling load

$$P_R == \frac{A\sigma_c}{1 + a\left(L_e/k\right)^2} = \frac{\left(88.75\pi\right)\left(335\right)}{1 + \left(\frac{1}{7500}\right)\left(\frac{2300}{12.6}\right)^2} = 17{,}160\ N$$

- ***Example 2-9****: The Figure shows a built-up column consisting of 150 mm x 100 mm R. S. J. with 120 mm x 12 mm plate riveted to each flange. Calculate the safe load, the column can carry, if it is 4 m long having one end fixed and the other hinged with a factor of safety 3.5. Take the properties of the joist as Area= 2167 mm^2 I_{XX} = 8.391 x 10^6 mm^4, I_{YY} = 0.948 x 10^6 mm^4. Assume the yield stress as 315 MPa and Rankine's constant (a)* = 1/7500



- **Solution**
- Given: Length of the column (*l*)= 4 m = 4 x 10^3 mm; Factor of safety = 3.5; Yield stress (σ_c) = 315.MPa = 315 N/mm^2; Area of joist= 2167 mm^2; Moment of inertia, about *X-X* axis *(I_{XX})* = 8.391 x 10^6 mm^4; Moment of inertia about *Y-Y* axis *(I_{YY})* = 0.948 x 10^6 mm^4 and Rankine's constant *(a)* = 1/7500

- From the geometry of the figure, we find that area of the column section

- A = 2167 + (2 x 120 x 12) = 5047 mm^2

- Moment of inertia of the column section bout *X-X* axis

$$I_{XX} = \left(83.91x10^6\right) + 2\left[\frac{(120)(12)^3}{12} - (120)(12)(81)^2\right] = 27.32x10^6\ mm^4$$

- Similarly $$I_{YY} = \left(0.948x10^6\right) + 2\left[\frac{(12)(120)^3}{12}\right] = 4.404x10^6\ mm^4$$

- Since I_{YY} is less than I_{XX} therefore the column will tend to buckle in *Y-Y* direction.

- Thus we shall take I equal to I_{YY} = 4.404 x 10^6 mm^4 *(i.e.,* least of two).

- Moreover as the column is fixed at one end and hinged at the other, therefore equivalent length of the column

$$L_e = \frac{l}{\sqrt{2}} = \frac{4000}{\sqrt{2}} = 2.83x10^3\ mm$$

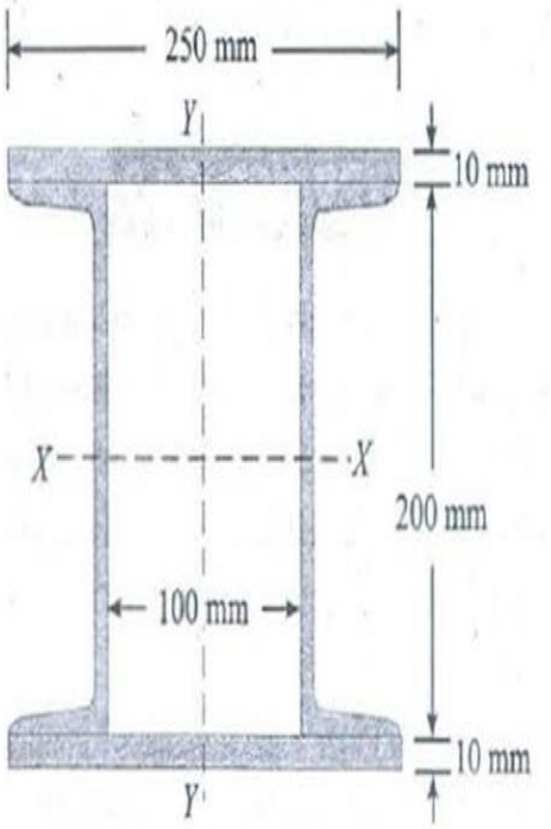
- The least radius of gyration $k = \sqrt{\frac{I}{A}} = \sqrt{\frac{4.404x10^6}{5047}} = 29.5\ mm$

- Therefore, Rankine's crippling load

$$P_R == \frac{A\sigma_c}{1 + a\left(L_e/k\right)^2} = \frac{\left(5047\right)\left(315\right)}{1 + \left(\frac{1}{7500}\right)\left(\frac{2830}{29.5}\right)^2} = 714\ kN$$

- *Example 2-10: A column is made up of two channels. ISJC 200 and two 250 mm* x *10 mm flange plates as shown. Determine by Rankine's formula the safe load, the column of 6 m length, with both ends fixed, can carry with a factor of safety 4. The properties of one channel are Area = I 777 mm^2, I_{XX} = 11.612* X *106 mm^4 and I_{YY} = 0.842* x *10^6 mm^4. Distance of centroid from back to web=. 19.7 mm. Take (σ_c) = 320 MPa and (a)= 1/*7500

- **Solution**
- Given : Length of the column (l)= 6 m = 6 X 10^3 mm ; Factor of safety= 4 ; Area of channel = 1777 mm^2; Moment of inertia about *X-X* axis (I_{XX}) = 11.612 x 10^6 mm^4; Moment of inertia about *Y-Y* axis (I_{YY}) = 0.842 x 10^6 mm^4; Distance of centroid from the back of web= 19.7 mm; Crushing stress (σ_c) = 320 MPa = 320 N/mm^2 and Rankine's constant (a)= 1/7500



- From the geometry of the figure, we find that area of the column section, A = 2 [1777 + (250 x 10)] = 8554 mm^2

- Moment of inertia of the column section about

$$I_{XX} = \left(2x11.612x10^6\right) + 2\left[\frac{(250)(10)^3}{12} - (250)(10)(105)^2\right] = 78.391x10^6\ mm^4$$

- Similarly $I_{YY} = 2\left[\frac{(10)(250)^3}{12} + \left(0.846x10^6\right) + 1777\,x\left(50 + 19.7\right)^2\right] = 44.992x10^6\ mm^4$

- Since I_{YY} is less than I_{XX} therefore the column will tend to buckle in *Y-Y* direction.

- Thus we shall take *I* equal to I_{YY} = 44.992 x 10^6 mm^4 *(i.e.,* least of two).

- Moreover as the column is fixed at one end and hinged at the other, therefore equivalent length of the column,

$$L_e = \frac{l}{2} = \frac{6000}{2} = 3x10^3\ mm$$

- The least radius of gyration

$$k = \sqrt{\frac{I}{A}} = \sqrt{\frac{44.992\,x10^6}{8554}} = 72.5\ mm$$

Therefore, Rankine's crippling load

$$P_R == \frac{A\sigma_c}{1 + a\left(L_e/k\right)^2} = \frac{(8554)(320)}{1 + \left(\frac{1}{7500}\right)\left(\frac{3000}{72.5}\right)^2} = 2228.5\ kN$$

- Safe load on the column

$$Safe\ load = \frac{Crippling\ load}{Factor\ of\ \ safety} = \frac{2228.5}{4} = 557.1\ kN$$

Case 3: Johnson's Formula for Columns

- Prof. Johnson, after a series of experiments and observations, proposed the following two formulae for columns:
  - Straight line formula and
  - Parabolic formula

Johnson's Straight Line Formula for Columns

- This formula was first proposed by Johnson, which states

$$P = A\left[\sigma_C - n\left(\frac{L_e}{k}\right)\right]$$

- where
- P Safe load on the column
- A Area of the column cross-section
- σ_C Allowable compressive stress in the column material
- n A constant, whose value depends upon the column material
- $\frac{L_e}{k}$ Slenderness ratio

- The values of σ_c and n are given in the following table

S. No.	Material	σ_c in MPa	n
1.	Mild Steel	320	0.0053
2.	Wrought Iron	250	0.0053
3.	Cast Iron	550	0.008

- A little consideration will show that for short columns shows that safe load $P = \sigma_C . A$

- But for long columns, there is always a possibility of buckling.

- It is thus obvious that the safe load will be small, depending upon the slenderness ratio.

Johnson's Parabolic Formula for Columns

- Prof. Johnson, after proposing the straight line formula, found that the results obtained by this formula are very approximate.

- He then proposed another formula, which states

$$P = A\left[\sigma_C - r\left(\frac{L_e}{k}\right)^2\right]$$

- where
- P Safe load on the column
- A Area of the column cross-section
- σ_c Allowable compressive stress in the column material
- r A constant, whose value depends upon the column material and
- $\frac{L_e}{k}$ Slenderness ratio

- The values of σ_c and r are given in the following table

S. No.	Material	σ_c in MPa	r
1.	Mild Steel	320	0.000057
2.	Wrought Iron	250	0.000039
3.	Cast Iron	550	0.000016

- **NOTE:**
- **Both the Johnson's straight line formula and Johnson's parabolic formula are rarely used these days**

- **They have been mentioned here only for their academic importance only.**

Case 4: Indian Standard Code for Columns

- The Bureau of Indian Standards (I. S. I.) has also given a code for the safe stress in I. S. 226-19621 which states

$$\sigma_C = \sigma'_C = \sigma_y / m \left[1 + 0.20 \sec\left(\frac{L_e}{k} \sqrt{\frac{m p'_c}{4E}} \right) \right]^{-1} \quad \text{for } \frac{L_e}{k} = 0 \text{ to } 160$$

$$\sigma_C = \sigma'_C \left(1.2 - \frac{L_e}{800k} \right) \quad \text{for } \frac{L_e}{k} = 160 \text{ and above}$$

where

- σ_c Allowable axial compressive stress
- σ'_c A value obtained from the above secant formula
- σ_y The guaranteed minimum yield stress

- m Factor of safety taken as 1.68
- L_e/k Slenderness ratio with equivalent column length
- E Modulus of elasticity equal to 200 GPa

- The I. S. I. has also given a table in l. S. 800 -1962 which gives the values of σ_c for mild steel for slenderness ratio from 0 to 350.

- The value of σ_y *i.e.,* the guaranteed minimum yield stress for mild steel is taken as 260 MPa. This table is given below:

$\frac{L}{k}$	σ_c in MPa	$\frac{L}{k}$	σ_c in MPa	$\frac{L}{k}$	σ_c in MPa
0	125	90	92.8	180	33.6
10	124.6	100	84.0	190	30.0
20	123.9	110	75.3	200	27.0
30	122.4	120	67.1	210	24.3
40	120.3	130	59.7	220	21.9
50	117.2	140	53.1	230	19.9
60	113.0	150	47.4	240	18.1
70	102.5	160	42.3	300	10.9
80	100.7	170	37.7	350	3.6

- **NOTE. Intermediate values may be obtained by linear interpolation**

- ***Example*** *2-11: A hollow cylindrical steel tube of 38 mm external diameter and* 2.5 *mm thick is used as a column of 2.3 m. long with both ends hinged. Determine the safe load by I. S. code.*

- **Solution**
- Given: External diameter (D) = 38 mm; Thickness = 2.5 mm and length of column (l) = 3 m = 3 x 10^3 mm.

- Area of the column section $A = \frac{\pi}{4}\left[D^2 - d^2\right] = \frac{\pi}{4}\left[(38)^2 - (33)^2\right] = 278.8\ mm^2$

- Moment of inertia of column section

$$I = \frac{\pi}{64}\left[D^4 - d^4\right] = \frac{\pi}{64}\left[(38)^4 - (33)^4\right] = 44.14x10^3\ mm^4$$

- The least radius of gyration $k = \sqrt{\frac{I}{A}} = \sqrt{\frac{44.14x10^3}{278.8}} = 12.6\ mm$

- Since the column is hinged at its both ends, therefore effective length of the column $L_e = l = 2300\ mm$

- Slenderness ratio $\frac{L_e}{k} = \frac{2300}{12.6} = 182.5$

- From table, we find that allowable stress for slenderness ratio of 182.5 is 32.7 MPa.

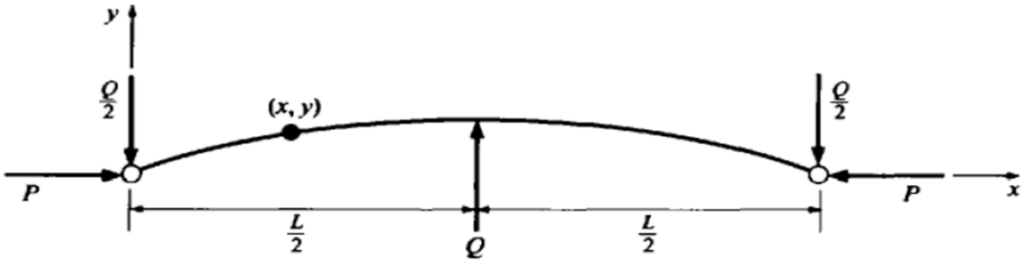
- Therefore safe load on the column

$$P = \sigma_C .A = (32.7)(278.8) = 9117 \ N$$

Assignment

1. Calculate the slenderness ratio of a strut made from a hollow tube 20 mm outside diameter and 16 mm inside diameter and 1.2 meters long.
2. Find the radius of gyration and the slenderness ratio of a strut made from 5 m length of hollow tube 50 mm outer diameter and 40 mm inner diameter.
3. Show that, for a pin – jointed Strut shown below, the maximum deflection is given by

$$\delta = \frac{Q}{2P\alpha}\tan\left(\frac{\alpha l}{2}\right) - \frac{Ql}{4P}$$



Assignment cont'd…

4. A rod of rectangular section 76 mm deep and 38 mm wide, is supported horizontally through pin joints at its ends, and carries a vertical load of 350 kg/m length and an axial thrust of 80 kN. If its length is 2.75 m, estimate the maximum stress induced. *E=206 GPa*